

Question 1

- (a) True
- (b) False
- (c) False
- (d) True

Question 2

First note that $33 = 3 \times 11$. By the Third Sylow Theorem, $n_3 \equiv 1 \pmod{3}$ and $n_3 | 11$ so $n_3 = 1$. Similarly, $n_{11} \equiv 1 \pmod{11}$ and $n_{11} | 3$. So $n_{11} = 1$. Let H be the unique Sylow 3-subgroup and K the unique Sylow 11-subgroup. Since their orders are prime, they are both cyclic and so $H \cong \mathbb{Z}_3$ and $K \cong \mathbb{Z}_{11}$. Because they are unique, $H \triangleleft G$ and $K \triangleleft G$. By Lagrange's Theorem, every non-identity element in H has order 3 and every non-identity element in K has order 11, so H and K are disjoint. To show that $G = HK$, we use the Second-Isomorphism Theorem. Since $H \leq G$ and $K \triangleleft G$, $H/(H \cap K) \cong (HK)/K$. But $|H \cap K| = 1$, so by Lagrange's Theorem, $|H/(H \cap K)| = \frac{|H|}{|H \cap K|} = |H| = \frac{|HK|}{|K|}$. That is, $|HK| = |H||K| = 33 = |G|$ and since $HK \leq G$ (from The Second Isomorphism Theorem), we get $HK = G$. Thus $G \cong H \times K \cong \mathbb{Z}_3 \times \mathbb{Z}_{11}$.